

Assignment 3: Understanding Customer Tipping Behavior

Dataset description

The dataset contains information about restaurant bills and tips. Each row represents one dining group.

- `total_bill` : total cost of the meal
- `tip` : tip amount
- `sex` : gender of the person paying
- `smoker` : whether the group included smokers
- `day` : day of the week
- `time` : lunch or dinner
- `size` : group size

Imports

```
In [3]: import pandas as pd
import seaborn as sns
```

Q1. Loading and Understanding the Dataset

1. Load the `tips` dataset from seaborn

```
In [49]: df = sns.load_dataset("tips")
```

2. Display the first 5 rows

```
In [50]: df.head()
```

```
Out[50]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

3. Print:

- Dataset shape
- Column names

```
In [51]: df.shape
```

```
Out[51]: (244, 7)
```

```
In [13]: df.columns
```

```
Out[13]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='str')
```

```
In [52]: # checking types of columns and null values
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex        244 non-null    category
3   smoker     244 non-null    category
4   day        244 non-null    category
5   time       244 non-null    category
6   size       244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

Q2. Basic Filtering - Who tips more?

1. Filter only weekend customers (Saturday & Sunday).

```
In [46]: df["day"].unique()
```

```
Out[46]: ['Sun', 'Sat', 'Thur', 'Fri']
Categories (4, str): ['Thur', 'Fri', 'Sat', 'Sun']
```

```
In [27]: weekend = df[df["day"].isin(('Sat', 'Sun'))]
```

```
In [28]: weekday = df[~df["day"].isin(('Sat', 'Sun'))]
```

2. Among them, find:

- Average tip
- Average total bill

```
In [34]: # 2 - weekend
weekend_mean_tip = weekend["tip"].mean()
weekend_mean_tip
```

```
Out[34]: np.float64(3.115276073619632)
```

```
In [36]: weekend_mean_total_bill = weekend["total_bill"].mean()
weekend_mean_total_bill
```

```
Out[36]: np.float64(20.893006134969326)
```

```
In [37]: # 2 - weekday
weekday_mean_tip = weekday["tip"].mean()
weekday_mean_tip
```

```
Out[37]: np.float64(2.7628395061728392)
```

```
In [38]: weekday_mean_total_bill = weekday["total_bill"].mean()
weekday_mean_total_bill
```

```
Out[38]: np.float64(17.55814814814815)
```

3. Compare this with weekday customers

Weekend customers leave more tips and their bills are higher.

```
In [47]: # Tips difference
weekend_mean_tip - weekday_mean_tip
```

```
Out[47]: np.float64(0.35243656744679264)
```

```
In [42]: # Total bills difference
weekend_mean_total_bill - weekday_mean_total_bill
```

```
Out[42]: np.float64(3.334857986821177)
```

Q3. Sorting and Identifying High-Value Customers

1. Sort customers by `total_bill` (descending)

```
In [44]: df_sorted = df.sort_values(by="total_bill", ascending=False)
```

```
In [45]: df_sorted.head()
```

```
Out[45]:
```

	total_bill	tip	sex	smoker	day	time	size
170	50.81	10.00	Male	Yes	Sat	Dinner	3
212	48.33	9.00	Male	No	Sat	Dinner	4
59	48.27	6.73	Male	No	Sat	Dinner	4
156	48.17	5.00	Male	No	Sun	Dinner	6
182	45.35	3.50	Male	Yes	Sun	Dinner	3

2. Display the **top 10 highest bills**

```
In [55]: df_sorted.iloc[:10]
```

```
Out[55]:
```

	total_bill	tip	sex	smoker	day	time	size
170	50.81	10.00	Male	Yes	Sat	Dinner	3
212	48.33	9.00	Male	No	Sat	Dinner	4
59	48.27	6.73	Male	No	Sat	Dinner	4
156	48.17	5.00	Male	No	Sun	Dinner	6
182	45.35	3.50	Male	Yes	Sun	Dinner	3
102	44.30	2.50	Female	Yes	Sat	Dinner	3
197	43.11	5.00	Female	Yes	Thur	Lunch	4
142	41.19	5.00	Male	No	Thur	Lunch	5
184	40.55	3.00	Male	Yes	Sun	Dinner	2
95	40.17	4.73	Male	Yes	Fri	Dinner	4

3. What do you notice about their group sizes?

Groups sizes `size` of top 10 highest bills are all higher than 1, ranging from 2 to 6

Q4. Data Quality Check & Cleaning

1. Check if the dataset has missing values

```
In [56]: df.isna().sum()
```

```
Out[56]: total_bill    0
tip                0
sex                0
smoker            0
day               0
time              0
size              0
dtype: int64
```

2. Why **missing values are dangerous** in machine learning?

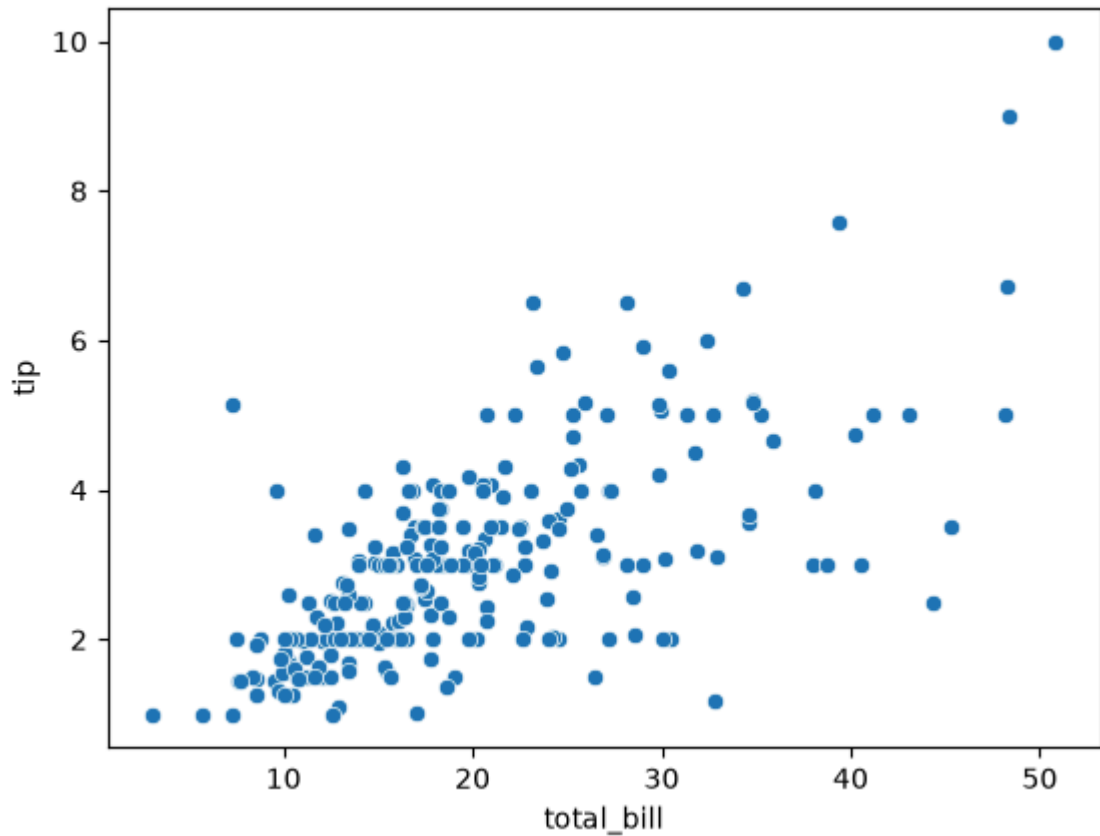
Because they can break internal algorithms, provide the bias in data, leading to false predictions and underestimating features significance

Q5. Visualization - What Influence Tips?

1. Plot
2. Interpret each plot in 1-2 sentences

- Tip vs Total Bill (scatter plot)

```
In [60]: sns.scatterplot(data=df, x="total_bill", y="tip");
```



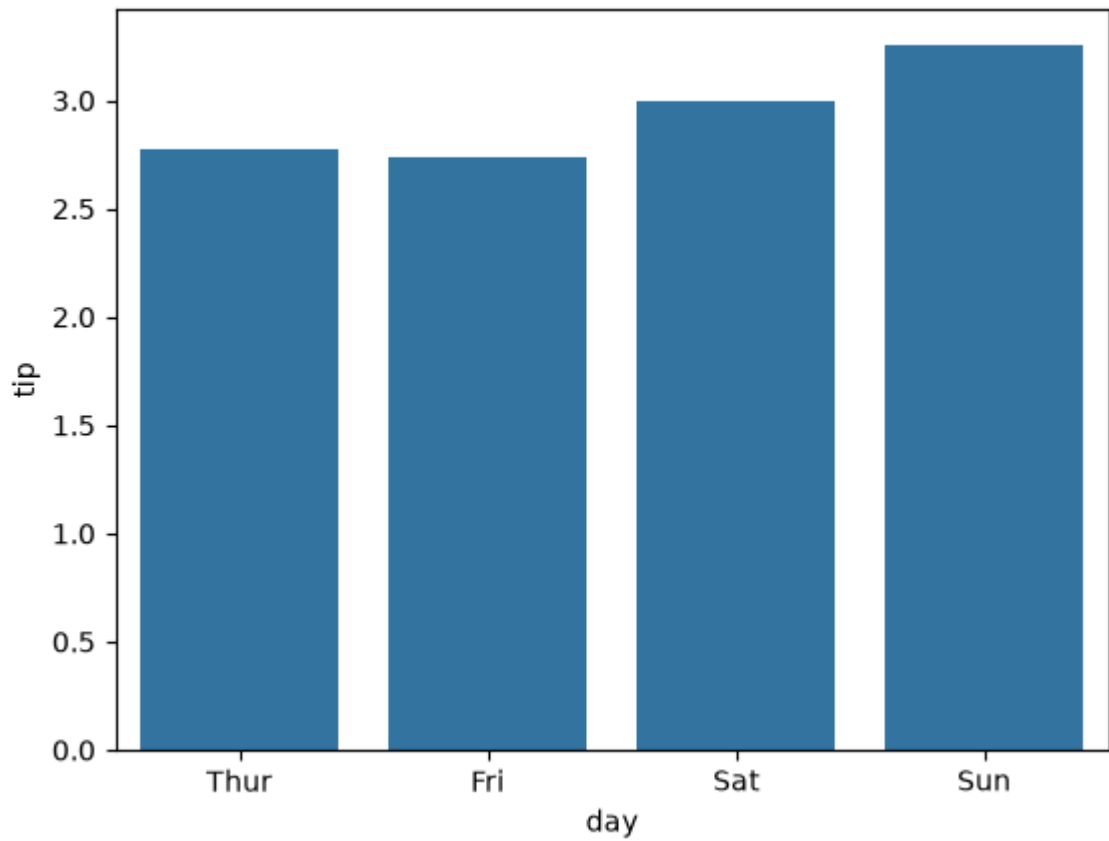
The scatterplot shows the relationship between total_bill and tip. From this scatterplot we can see the positive linear correlation between those 2 variables (tip is increasing with total_bill in a straight line)

- Average tip by day (bar plot)

```
In [64]: df.groupby(by="day")["tip"].mean()
```

```
Out[64]: day
Thur    2.771452
Fri     2.734737
Sat     2.993103
Sun     3.255132
Name: tip, dtype: float64
```

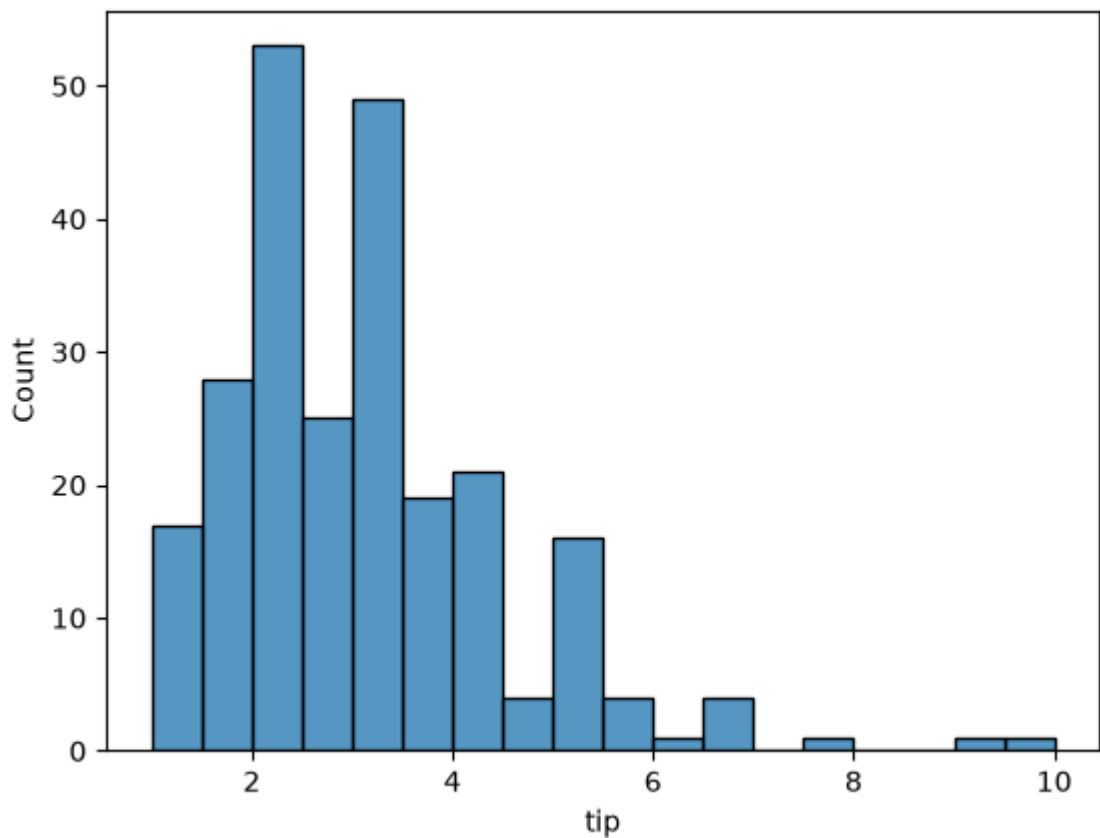
```
In [70]: sns.barplot(data=df.groupby(by="day")["tip"].mean());
```



Barplot shows the data of average tip accross distinct categories, week days. Weekends have the highest average tip.

- Tip distribution (histogram)

```
In [72]: sns.histplot(data=df, x="tip", binwidth=0.5);
```



The histogram shows the distribution of all tips made, with single bin width of 0.5. Tips from 2 to 2.5 and from 3 to 3.5 are the most frequent.

Q6. Final Insight & Conclusion

- Who tips more?

```
In [75]: df.groupby(by="sex")["tip"].mean()
```

```
Out[75]: sex
Male      3.089618
Female    2.833448
Name: tip, dtype: float64
```

```
In [76]: df.groupby(by="smoker")["tip"].mean()
```

```
Out[76]: smoker
Yes      3.008710
No       2.991854
Name: tip, dtype: float64
```

```
In [77]: df.groupby(by="size")["tip"].mean()
```

```
Out[77]: size
1      1.437500
2      2.582308
3      3.393158
4      4.135405
5      4.028000
6      5.225000
Name: tip, dtype: float64
```

There were no question that could help answer if there is some group of people of specific gender or with smoking habits that tips more. Though I calculated average tips based on the `sex`, `smoker`, and `size` attributes.

There was no difference between smokers and non-smokers, but males tipped higher compared to females and the larger the group size, the higher the tip.

- When tips are highest?
 1. The tips are highest on the weekends, Saturday and Sunday.
 2. The higher the bill, the higher the tip.
- How this insight could help restaurant management?

Based on certain insights from the analysis, the restaurant management team can develop their plan of actions in order to increase the business income by preferring certain customers over others. Apart that, I don't see any useful information, that can help develop business.